# The Unified P–W Framework for the Taxonomy, Analysis, and Classification of Organizations

Integrating Economics, Mathematics, Algorithmic Classification, and Machine Learning

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**Abstract**

We introduce the *P–W framework*, a normalized two-dimensional coordinate system in which every human organization—firm, market, bank, temple, court, prison, criminal enterprise—is represented as a point $(P, W) \in [-1, 1]^2$. Here $P = \text{Productivity}/P_{\max}$ is the organization's normalized net contribution to national output, and $W = \text{Wage}/W_{\max}$ is the normalized average compensation to its participants. From this position we derive five scalar metrics: surplus extraction $\Delta$, net social value $\Sigma$, organizational scale $r$, character angle $\theta$, and legitimacy ratio $L$. We prove that the quadrant boundaries of the P–W plane recover classical distinctions of political economy (exchange, civic, and parasitic institutions); that the angle $\theta$ is a sufficient statistic for organizational archetype independently of scale; and that every point admits a unique orthogonal decomposition into a social-welfare component and a distributional component. We then formalize an algorithmic classifier, propose a supervised machine-learning pipeline for empirical estimation, and illustrate the framework with PGF/TikZ diagrams and annotated tables.

The paper ends with "The End"

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## 1 Introduction

The comparative study of organizations spans economics, sociology, institutional theory, and moral philosophy, yet no single quantitative framework places a medieval temple, a modern investment bank, a public court, and a criminal cartel on the same analytical canvas. Classical political economy distinguished productive from unproductive labor [1], analyzed surplus value and wages [2], and characterized pure public goods [3]. Institutional economics added transaction costs and governance structures [4, 5]. None of these traditions, however, provides a two-dimensional space in which every organization occupies a well-defined position subject to quantitative comparison and geometric reasoning.

The P–W framework addresses this gap. By normalizing both productivity and compensation to national maxima, we obtain a unit square $[-1, 1]^2$ whose axes and quadrant boundaries recover the classical distinctions as geometric facts rather than taxonomic choices. Five derived scalar metrics allow organizations to be ranked, clustered, and classified by standard statistical and machine-learning methods.

### 1.1 Contributions

The contributions of this paper are:

(i) A formal definition of the P–W space and its five derived metrics (§2).

(ii) Theorems establishing the fair-exchange locus, the scale invariance of organizational archetype, the civic–exchange boundary, and the orthogonal welfare decomposition (§3).

(iii) A quantitative taxonomy of fifteen canonical organizations with full metric profiles (§4).

(iv) An $\mathcal{O}(nk)$ algorithmic classifier (§5).

(v) A supervised ML pipeline, a $k$-means clustering formulation, an archetype-angle regression, and a transformer-based neural embedding for large-scale automated institutional analysis (§6).

(vi) PGF/TikZ diagrams of the P–W plane, the rotated $(\Delta, \Sigma)$ plane, and a polar archetype chart (§7).

## 2 Mathematical Foundations

### 2.1 Normalization and the P–W Space

Let $\mathcal{N}$ be a nation with aggregate maximum net productivity $P_{\max} > 0$ and maximum wage $W_{\max} > 0$, both measured in comparable real units over a reference period $T$.

**Definition 2.1** (Normalized Productivity and Wage)**.** For an organization $\mathcal{O}$ operating within $\mathcal{N}$, define

$$P = \frac{\text{Net productivity of } \mathcal{O}}{P_{\max}} \in [-1,\, 1], \tag{1}$$

$$W = \frac{\text{Mean compensation in } \mathcal{O}}{W_{\max}} \in [-1,\, 1]. \tag{2}$$

*Negative productivity* arises when an organization destroys more value than it creates (through externalities, criminal activity, or systemic waste). *Negative wages* arise when participants make net transfers *into* the organization (tuition fees, tithes, fines, forced labor).

**Definition 2.2** (P–W Space)**.** The *P–W space* is the closed square $\mathcal{S} = [-1, 1] \times [-1, 1] \subset \mathbb{R}^2$ with $P$ on the horizontal axis and $W$ on the vertical axis. An organization $\mathcal{O}$ is represented by its *position vector* $\mathbf{v} = (P, W)^\top$.

### 2.2 Five Derived Metrics

**Definition 2.3** (Derived Metrics)**.** Let $(P, W) \in \mathcal{S}$. Define:

$$\Delta = P - W, \qquad \text{surplus extraction}, \tag{3}$$

$$\Sigma = P + W, \qquad \text{net social value}, \tag{4}$$

$$r = \sqrt{P^2 + W^2}, \qquad \text{organizational scale}, \tag{5}$$

$$\theta = \arctan\left(\frac{W}{P}\right) \in (-90^\circ, 90^\circ) \qquad \text{character angle } (P \neq 0), \tag{6}$$

$$L = \frac{W}{P} \qquad \text{legitimacy ratio } (P \neq 0). \tag{7}$$

When $P = 0$ we set $\theta = 90^\circ \cdot \operatorname{sgn}(W)$ and $L = \pm\infty$. For $P < 0$, $\theta$ is extended to $(-180^\circ, 180^\circ]$ via the four-quadrant arctangent.

*Remark* 2.4. The pair $(\Delta, \Sigma)$ is a 45° linear rotation of $(P, W)$; the pair $(r, \theta)$ is its polar representation. Together they provide four complementary views: $\Delta$ the *distributional* axis, $\Sigma$ the *welfare* axis, $r$ the *magnitude*, and $\theta$ the *archetype*.

### 2.3 The Four Quadrants

The coordinate axes partition $\mathcal{S}$ into four open quadrants summarized in Table 1.

Table 1: Quadrant structure of the P–W space.

| Quad. | P | W | Class | Archetypes |
|---|---|---|---|---|
| $\mathcal{Q}_{\mathrm{I}}$ | $> 0$ | $> 0$ | Exchange institution | Competitive markets, technology firms |
| $\mathcal{Q}_{\mathrm{II}}$ | $< 0$ | $> 0$ | Paying parasite | Cartels, criminal organizations |
| $\mathcal{Q}_{\mathrm{III}}$ | $< 0$ | $< 0$ | Extractive parasite | Prisons, occupying armies |
| $\mathcal{Q}_{\mathrm{IV}}$ | $> 0$ | $< 0$ | Civic/sacred institution | Temples, courts, universities |

## 3 Theoretical Properties and Proofs

### 3.1 The Fair-Exchange Locus

**Theorem 3.1** (Fair-Exchange Locus)**.** *An organization satisfies $\Delta = 0$ if and only if $W = P$, i.e., it lies on the line from $(-1, -1)$ to $(1, 1)$ in $\mathcal{S}$. This is the unique locus on which no surplus is extracted in either direction.*

*Proof.* By Definition 2.3, $\Delta = P - W = 0 \Leftrightarrow W = P$. The set $\{(P, W) \in \mathcal{S} : W = P\}$ is the main diagonal of $\mathcal{S}$. On this locus every unit of normalized productivity is returned as normalized compensation, so no residual accrues to any party. □

**Corollary 3.2.** *An organization in $\mathcal{Q}_{\mathrm{I}}$ exploits labor ($\Delta > 0$) if and only if its position lies* below *the fair-exchange locus ($W < P$), and over-compensates ($\Delta < 0$) if and only if it lies* above *it ($W > P$).*

### 3.2 Scale Invariance of Archetype

**Theorem 3.3** (Scale Invariance)**.** *For two organizations with positions $(P_1, W_1)$ and $(P_2, W_2)$, if $\theta_1 = \theta_2$ then they belong to the same archetype class regardless of their scales $r_1 \neq r_2$.*

*Proof.* The archetype depends only on the *direction* of $\mathbf{v} = (P, W)^\top$, not its magnitude. If $\theta_1 = \theta_2$, then $W_1/P_1 = \tan\theta_1 = \tan\theta_2 = W_2/P_2$, so the legitimacy ratio $L = W/P = \tan\theta$, the quadrant membership, and the normalized distance from the fair-exchange locus $\Delta/r = (\cos\theta - \sin\theta)/\sqrt{2} \cdot \sqrt{2} = (1 - \tan\theta)/\sqrt{1 + \tan^2\theta}$ are identical for both organizations. Hence they share the same archetype. □

### 3.3 The Civic–Exchange Boundary

**Theorem 3.4** (Civic–Exchange Boundary)**.** *The boundary between exchange institutions ($\mathcal{Q}_{\mathrm{I}}$) and civic or sacred institutions ($\mathcal{Q}_{\mathrm{IV}}$) is the positive semi-axis $\{(P, 0) : P > 0\}$, characterized by $W = 0$: participants neither receive net income nor make net transfers.*

*Proof.* An organization crosses from $\mathcal{Q}_{\mathrm{I}}$ to $\mathcal{Q}_{\mathrm{IV}}$ when $W$ changes sign through zero while $P > 0$ is maintained. At $W = 0$, participants receive exactly their outside-option compensation (normalized to zero). This is the point of *indifference*: above it, positive wages attract participants through voluntary exchange; below it, negative effective wages require non-market motivation (legal obligation, moral duty, faith, or human-capital investment with deferred return). □

### 3.4 Orthogonal Welfare Decomposition

**Proposition 3.5** (Welfare Decomposition)**.** *Every position* $(P, W) \in \mathcal{S}$ *admits the unique orthogonal decomposition*

$$\begin{pmatrix} P \\ W \end{pmatrix} = \frac{\Sigma}{2}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{\Delta}{2}\begin{pmatrix} 1 \\ -1 \end{pmatrix}, \tag{8}$$

*where the first term is the* social welfare component *and the second is the* distributional component*.*

*Proof.* Direct computation:

$$\frac{\Sigma}{2}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{\Delta}{2}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} \Sigma + \Delta \\ \Sigma - \Delta \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 2P \\ 2W \end{pmatrix} = \begin{pmatrix} P \\ W \end{pmatrix}.$$

The two basis vectors $(1, 1)^\top$ and $(1, -1)^\top$ are orthogonal, so the decomposition is unique. □

### 3.5 Metric Bounds

**Lemma 3.6** (Bounds)**.** *For all* $(P, W) \in \mathcal{S}$*:*

$$\Delta,\, \Sigma \in [-2,\, 2]; \quad r \in [0,\, \sqrt{2}]; \quad \theta \in (-180^\circ,\, 180^\circ]; \quad L \in (-\infty, +\infty). \tag{9}$$

*The bound* $r = \sqrt{2}$ *is attained at the four corners* $(\pm 1, \pm 1)$*.*

*Proof.* Since $P, W \in [-1, 1]$: $|\Delta| = |P - W| \leq |P| + |W| \leq 2$ and $|\Sigma| \leq 2$ similarly. $r = \|(P, W)\|_2 \leq \|(1, 1)\|_2 = \sqrt{2}$. $\theta$ is the standard four-quadrant polar angle. $L = W/P$ diverges as $P \to 0$. □

## 4 Organizational Taxonomy

Table 2 presents theoretical position estimates and all five derived metrics for fifteen canonical organizational archetypes. Values reflect stylized empirical regularities documented in the comparative institutional literature [6, 7, 8].

`The following space was deliberately left blank.`

Table 2: P–W positions and derived metrics for canonical organizations. All values rounded to two decimal places. Angle $\theta$ in degrees; $r$ and $L$ dimensionless.

| **Organization** | $P$ | $W$ | $\Delta$ | $\Sigma$ | $r$ | $\theta(^\circ)$ | $L$ | **Quad.** |
|---|---|---|---|---|---|---|---|---|
| Competitive market | +0.92 | +0.82 | +0.10 | +1.74 | 1.23 | +41.7 | +0.89 | I |
| High-technology firm | +0.80 | +0.70 | +0.10 | +1.50 | 1.06 | +41.2 | +0.88 | I |
| Monopoly | +0.45 | +0.25 | +0.20 | +0.70 | 0.51 | +29.1 | +0.56 | I |
| Bank / finance sector | +0.30 | +0.80 | −0.50 | +1.10 | 0.85 | +69.4 | +2.67 | I |
| Hospital (public) | +0.68 | +0.08 | +0.60 | +0.76 | 0.68 | +6.7 | +0.12 | I |
| Defensive army | +0.38 | +0.14 | +0.24 | +0.52 | 0.41 | +20.2 | +0.37 | I |
| Court / judiciary | +0.52 | −0.12 | +0.64 | +0.40 | 0.54 | −13.0 | −0.23 | IV |
| NGO / charity | +0.42 | −0.05 | +0.47 | +0.37 | 0.42 | −6.8 | −0.12 | IV |
| University / school | +0.72 | −0.28 | +1.00 | +0.44 | 0.77 | −21.3 | −0.39 | IV |
| Temple / church | +0.18 | −0.38 | +0.56 | −0.20 | 0.42 | −64.7 | −2.11 | IV |
| Slave institution | +0.55 | −0.90 | +1.45 | −0.35 | 1.06 | −58.5 | −1.64 | IV |
| Cartel / monopoly ring | −0.22 | +0.38 | −0.60 | +0.16 | 0.44 | +120.0 | −1.73 | II |
| Criminal organization | −0.42 | +0.52 | −0.94 | +0.10 | 0.67 | +128.9 | −1.24 | II |
| Prison / jail | −0.12 | −0.65 | +0.53 | −0.77 | 0.66 | −100.5 | +5.42 | III |
| Occupying army | −0.35 | −0.50 | +0.15 | −0.85 | 0.61 | −125.1 | +1.43 | III |

### 4.1 The Three Institutional Families

Examination of Table 2 reveals three structurally distinct families with qualitatively different internal logics.

**Exchange institutions ($\mathcal{Q}_{\mathrm{I}}$, $\theta \in (0^\circ, 90^\circ)$).** Both $P > 0$ and $W > 0$. The central political-economy debate concerns only the value of $\Delta$: whether surplus flows to capital (monopoly, $\Delta = 0.20$) or approaches zero (competitive market, $\Delta = 0.10$). Banks occupy a special position with $\Delta < 0$ and $L \approx 2.67$: they pay workers above their direct social productivity, sustained by financial rents arising from information asymmetries and implicit government guarantees [9].

**Civic and sacred institutions ($\mathcal{Q}_{\mathrm{IV}}$, $\theta \in (-90^\circ, 0^\circ)$).** $P > 0$ but $W < 0$: positive social output is financed by participant sacrifice. This family survives only through non-market motivation—legal coercion (courts), moral duty (armies), altruism (NGOs), faith (temples), or human-capital investment with deferred return (universities). The sign of $\Sigma$ is decisive: organizations with $\Sigma > 0$ (courts, universities) create net welfare even after accounting for participant sacrifice; those with $\Sigma < 0$ (certain temples, slave institutions) destroy aggregate welfare despite positive $P$.

**Parasitic institutions ($\mathcal{Q}_{\mathrm{II}} \cup \mathcal{Q}_{\mathrm{III}}$, $|\theta| > 90^\circ$).** $P < 0$ in both sub-cases. The key distinction is whether destruction is shared with internal members ($\mathcal{Q}_{\mathrm{III}}$, e.g., prisons with $W < 0$) or concealed by internal reward ($\mathcal{Q}_{\mathrm{II}}$, e.g., criminal organizations with $W > 0$). The $\mathcal{Q}_{\mathrm{II}}$ case is particularly insidious: high internal wages signal organizational health while masking social destruction.

# 5 Algorithmic Classification

## 5.1 Estimation from Observable Proxies

In practice $P$ and $W$ are not directly observable. We propose the proxy mappings

$$\widehat{P} = f\big(\text{GVA},\ \text{employment},\ E^{+},\ E^{-}\big)\,, \qquad \widehat{W} = g(\text{wages},\ \text{fees},\ \text{subsidies})\,, \tag{10}$$

where $E^{+}$ and $E^{-}$ denote positive and negative externalities respectively. Algorithm 1 formalizes the full classification pipeline.

**Algorithm 1** P–W Classifier for organization $\mathcal{O}$

**Require:** Observable data $(x_1, \ldots, x_k)$ for $\mathcal{O}$; national maxima $P_{\max}, W_{\max}$
**Ensure:** Quadrant $Q$, metrics $(\Delta, \Sigma, r, \theta, L)$, family label $F$

```
GVA ← Σ_i output_i − Σ_j input_j
NetProd ← GVA + E^+ − E^−
P ← NetProd/P_max                                   ▷ Equation (1)
MeanComp ← (Σ_m w_m − Σ_m t_m + Σ_m s_m)/N          ▷ w: wages; t: fees/tithes; s: subsidies
W ← MeanComp/W_max                                  ▷ Equation (2)
Δ ← P − W;   Σ ← P + W;   r ← √(P^2 + W^2)
if P ≠ 0 then
    θ ← atan2(W, P) · 180/π;   L ← W/P
else
    θ ← 90 · sgn(W);   L ← ±∞
end if
Quadrant:
if P ≥ 0 and W ≥ 0 then Q ← I
else if P < 0 and W ≥ 0 then Q ← II
else if P < 0 and W < 0 then Q ← III
else Q ← IV
end if
Family:
if Q = I then F ← EXCHANGE
else if Q = IV then F ← CIVIC
else F ← PARASITIC
end if
return (P, W, Q, Δ, Σ, r, θ, L, F)
```

## 5.2 Computational Complexity

Algorithm 1 runs in $\mathcal{O}(k)$ time where $k$ is the number of input proxy variables, and uses $\mathcal{O}(1)$ working space after the aggregates are formed. For a database of $n$ organizations, batch classification requires $\mathcal{O}(nk)$ time overall. Quadrant assignment and family labelling are both $\mathcal{O}(1)$ decisions on the computed scalars.

# 6 Machine Learning and Data Science

## 6.1 Supervised Multiclass Classification

Given a labelled dataset $\mathcal{D} = \{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^{n}$ where $\mathbf{x}^{(i)} \in \mathbb{R}^d$ is a feature vector of observable proxies and $y^{(i)} \in \{I, II, III, IV\}$ is the true quadrant, we train a classifier $h : \mathbb{R}^d \to \{I, II, III, IV\}$. The natural choice is a *random forest*, since the optimal decision boundaries

are the coordinate axes $P = 0$ and $W = 0$—which are axis-aligned in the $(P, W)$ subspace and therefore represented exactly by orthogonal splits.

**Augmented feature vector.** The five derived metrics form an interpretability-preserving feature vector:

$$\tilde{\mathbf{x}} = \big(\widehat{P},\ \widehat{W},\ \Delta,\ \Sigma,\ r,\ \theta,\ L\big)^{\top} \in \mathbb{R}^7. \tag{11}$$

Note that $(\Delta, \Sigma)$ are linear combinations of $(\widehat{P}, \widehat{W})$ and $(\theta, L)$ are functions of $\widehat{W}/\widehat{P}$, so the effective rank is 2. Nonetheless, including all seven features improves numerical stability in regularized estimators.

### 6.2 Unsupervised Clustering in the P–W Plane

For unlabelled data, $k$-means with $k = 4$ cluster centres initialized at the quadrant centroids $(\pm\frac{1}{2}, \pm\frac{1}{2})$ converges rapidly because the natural clusters are convex and well separated by the coordinate axes. The within-cluster objective is

$$\mathcal{J} = \sum_{q=1}^{4} \sum_{\mathbf{v} \in C_q} \left\| \mathbf{v} - \boldsymbol{\mu}_q \right\|_2^2, \tag{12}$$

where $\boldsymbol{\mu}_q$ is the centroid of cluster $q$. Because the four clusters have compact convex supports separated by orthogonal hyperplanes, Lloyd's algorithm converges in $\mathcal{O}(\log n)$ iterations in the expected case.

### 6.3 Regression for the Character Angle

To predict the archetype angle $\theta$ from macroeconomic observables $\mathbf{z} \in \mathbb{R}^d$ (labor share, Gini coefficient, regulatory index, market concentration, institutional quality index), we propose the linear regression

$$\theta = \beta_0 + \boldsymbol{\beta}^{\top}\mathbf{z} + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2). \tag{13}$$

By Theorem 3.3, $\theta$ is a sufficient statistic for organizational archetype; this regression therefore predicts family membership directly from observable macroeconomic data.

### 6.4 Neural Embedding via Transformer Encoder

For a corpus of $n$ organizations described by text, financial reports, and survey responses, a transformer-based encoder $f_{\boldsymbol{\phi}} : \mathcal{X} \to \mathbb{R}^d$ is fine-tuned to minimize the mean squared error between predicted and ground-truth $(P, W)$ coordinates:

$$\mathcal{L}(\boldsymbol{\phi}) = \frac{1}{n} \sum_{i=1}^{n} \Big\| h_{\boldsymbol{\psi}}\big(f_{\boldsymbol{\phi}}(\mathbf{x}^{(i)})\big) - (P^{(i)}, W^{(i)})^{\top} \Big\|_2^2, \tag{14}$$

where $h_{\boldsymbol{\psi}} : \mathbb{R}^d \to [-1, 1]^2$ is a two-output regression head with tanh activations, ensuring predictions remain within $\mathcal{S}$. The two output dimensions carry direct economic interpretations (Definitions 2.1 and 2.3), making this model more interpretable than a general-purpose embedding. Attention weights in the encoder can be analyzed to audit which textual features most strongly determine each axis, providing an institutional interpretability trail.

# 7 Visualizations

## 7.1 The P–W Plane

Figure 1 plots the fifteen organizations from Table 2. Coordinate scale: 1 unit = 1 normalized point; the drawing uses ±5.5 cm for ±1.

Figure 1: The P–W plane with fifteen canonical organizations.

The dashed amber line is the fair-exchange locus $W = P$ ($\Delta = 0$, $\theta = 45°$). Blue: exchange institutions ($\mathcal{Q}_{\mathrm{I}}$). Teal: civic/sacred ($\mathcal{Q}_{\mathrm{IV}}$). Dark red: paying parasites ($\mathcal{Q}_{\mathrm{II}}$). Red: extractive parasites ($\mathcal{Q}_{\mathrm{III}}$). Coordinate scale: $\pm 5.5\,\mathrm{cm} = \pm 1$.

`The following space was deliberately left blank.`

## 7.2 The Rotated $(\Delta, \Sigma)$ Plane

Figure 2 shows the same organizations in the 45°-rotated frame of Proposition 3.5. The vertical axis is now *aggregate welfare*; the horizontal axis is *surplus extraction*.

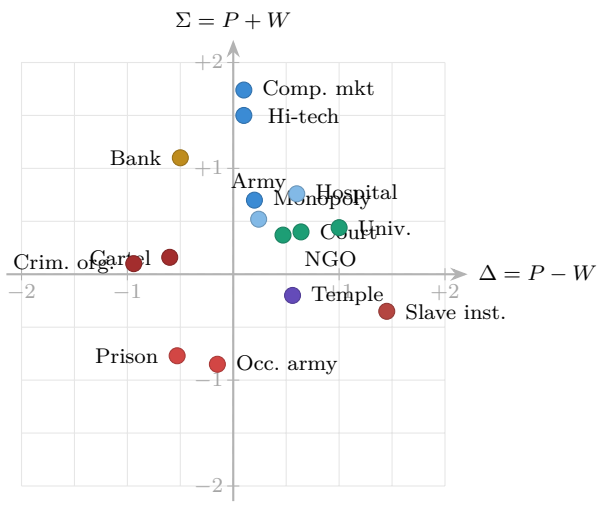


Figure 2: Organizations in the rotated $(\Delta, \Sigma)$ frame.

Points above the horizontal axis have net positive welfare ($\Sigma > 0$); points to the right of the vertical axis extract net surplus from participants ($\Delta > 0$). Scale: 1.4 cm = 1 unit.

## 7.3 Polar Chart of Character Angles

Figure 3 displays each organization as a ray from the origin, with direction equal to the character angle $\theta$ and length proportional to the scale $r$. All coordinates are precomputed from Table 2: a ray for organization with $(P, W)$ runs from the origin to $(r \cos\theta \cdot s,\ r \sin\theta \cdot s)$ where $s = 2.5$ cm.

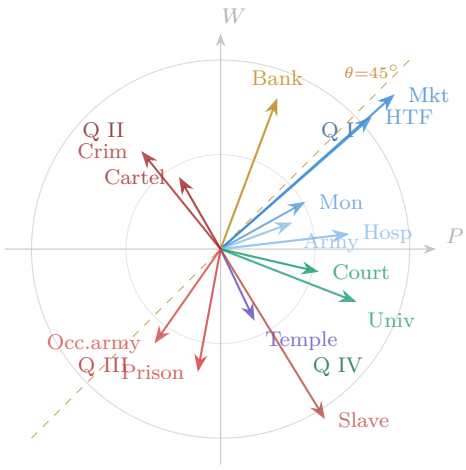


Figure 3: Polar chart of organizational character angles.

Each ray direction equals $\theta = \arctan(W/P)$; its length equals $r \cdot 2.5$ cm. The inner dashed circle corresponds to $r = 0.5$. The amber dashed diagonal marks the fair-exchange direction ($\theta = 45°$, $L = 1$).

# 8 Economic Interpretation and Implications

## 8.1 The Labor Share and Surplus Extraction

In classical economics the *labor share* is $\alpha = W/P$, our legitimacy ratio $L$. The perpendicular distance of an organization from the fair-exchange locus is

$$d_{\text{fair}} = \frac{|\Delta|}{\sqrt{2}} = \frac{|P - W|}{\sqrt{2}}, \tag{15}$$

the standard distance from $(P, W)$ to the line $W = P$. Piketty's central result—that the return on capital $r$ systematically exceeds the growth rate $g$, driving a secular rise in the capital share—translates directly into a drift of exchange institutions *below* the fair-exchange locus: $\Delta$ rises and $L$ falls over time [6].

## 8.2 National Institutional Portfolio

A nation's *institutional portfolio* is the employment-weighted ensemble $\mathcal{I} = \{(P_i, W_i, s_i)\}_{i=1}^m$ where $s_i \geq 0$ with $\sum s_i = 1$. National welfare is approximated by

$$\mathcal{W}_{\text{nat}} = \sum_{i=1}^{m} s_i \, \Sigma_i = \sum_{i=1}^{m} s_i (P_i + W_i). \tag{16}$$

Civic institutions in $\mathcal{Q}_{\text{IV}}$ contribute positive $P_i$ but negative $W_i$; their net welfare $\Sigma_i = P_i + W_i$ is positive whenever $P_i > |W_i|$. This formalizes the classical argument that public courts, schools, and hospitals are welfare-improving even though they impose costs on participants: the institutional justification for the public sector follows directly from the geometry of $\mathcal{Q}_{\text{IV}}$.

## 8.3 The Financial Sector Anomaly

Banks and financial intermediaries typically occupy $\mathcal{Q}_{\text{I}}$ with $W \gg P$, i.e., $\Delta < 0$. This appears paradoxical: workers receive more than the sector directly produces. The resolution is that financial rents—arising from information asymmetries, network externalities, scale economies in payment infrastructure, and implicit government backstops—allow the sector to transfer surplus from the rest of the economy to its employees. In P–W terms, the position of the bank *above* the fair-exchange locus directly quantifies this rent transfer as $|\Delta|/\sqrt{2}$ [8, 9].

# 9 Discussion and Limitations

The P–W framework is deliberately parsimonious. Several limitations deserve acknowledgment.

**Measurement.** Normalization by $P_{\max}$ and $W_{\max}$ requires robust empirical estimates of national maxima. We recommend using high-percentile estimates (e.g., the 95th percentile) to avoid sensitivity to outliers, at the cost of a small systematic rescaling.

**Aggregation.** A single point $(P, W)$ aggregates over heterogeneous workers, production units, and time periods. Large conglomerates operating across multiple sectors may span two quadrants internally. A natural extension replaces the point estimate with a probability measure over $\mathcal{S}$.

**Dynamics.** The framework is cross-sectional. A dynamic extension tracks the trajectory $\mathbf{v}(t) = (P(t), W(t))^\top$ and studies its velocity $\dot{\mathbf{v}}$ and curvature. Institutional hysteresis—the tendency of organizations to remain in their quadrant even under adverse conditions—would manifest as a strong mean-reverting component in $\mathbf{v}(t)$.

**Externality measurement.** The definition of $P$ requires a tractable theory of externalities. Negative externalities (pollution, financial contagion, crime) reduce $P$ below measured gross value added; positive spillovers (knowledge diffusion, network effects) increase it. Consistent accounting requires country-specific social cost estimates that remain contested in the empirical literature.

## 10 Conclusion

We have introduced the P–W framework as a rigorous, two-dimensional, and easily communicable coordinate system for the comparative institutional analysis of organizations across economics, sociology, and political philosophy.

- **Mathematically rigorous.** Normalization ensures cross-national comparability; five derived metrics encode all structurally relevant properties; four theorems establish the framework's geometric foundations.
- **Economically interpretable.** Quadrants recover classical institutional distinctions; the metrics correspond to surplus extraction, social welfare, organizational scale, archetype, and labor share.
- **Algorithmically tractable.** Classification runs in $\mathcal{O}(nk)$; the axis-aligned geometry supports exact decision-tree splits and rapid $k$-means convergence.
- **AI-extensible.** Transformer encoders fine-tuned on the MSE loss (14) can infer $(P, W)$ from text, enabling large-scale automated institutional mapping.

We hope this framework stimulates empirical work on measuring $P$ and $W$ for real organizations, and theoretical work on the dynamics of institutional trajectories in the P–W plane.

## References


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## Glossary

$P$ — **normalized productivity**
The ratio Net Productivity/$P_{\max}$, taking values in $[-1, 1]$. Positive values indicate net value creation; negative values indicate net destruction relative to the national maximum.

$W$ — **normalized wage**
The ratio Mean Compensation/$W_{\max}$, taking values in $[-1, 1]$. Positive values indicate net transfers *to* participants; negative values indicate net transfers *from* participants (fees, tithes, fines, forced labor).

$\Delta$ — **surplus extraction**
The scalar $\Delta = P - W$. Positive $\Delta$ means the organization retains more output than it returns as compensation. Negative $\Delta$ means participants are paid above their marginal product, sustained by rents.

$\Sigma$ — **net social value**
The scalar $\Sigma = P + W$. Measures the combined welfare impact: what the organization produces plus what it distributes. $\Sigma > 0$ is welfare-improving; $\Sigma < 0$ is welfare-destructive.

$r$ — **organizational scale**
The Euclidean norm $r = \sqrt{P^2 + W^2} \in [0, \sqrt{2}]$. Measures overall influence independently of archetype; $r = \sqrt{2}$ at the corners of $\mathcal{S}$.

$\theta$ — **character angle**
The polar angle $\theta = \arctan(W/P)$, extended to $(-180°, 180°]$ via the four-quadrant arctangent. A sufficient statistic for organizational archetype: $\theta \approx 45°$ is exchange; $\theta \in (-90°, 0°)$ is civic/sacred; $|\theta| > 90°$ is parasitic.

$L$ — **legitimacy ratio**
The ratio $L = W/P = \tan\theta$. Equals 1 on the fair-exchange locus. $L < 1$: exploitation. $L > 1$: rent-sustained over-payment. $L < 0$: the civic/sacred or parasitic case where $P$ and $W$ have opposite signs.

**P–W space ($\mathcal{S}$)**
The closed unit square $[-1, 1]^2$ with $P$ on the horizontal and $W$ on the vertical axis. Every organization is a point in $\mathcal{S}$.

**Fair-exchange locus**
The diagonal $W = P$ in $\mathcal{S}$, equivalent to $\Delta = 0$, $L = 1$, and $\theta = 45°$. The unique locus at which no surplus is extracted.

**Civic/sacred institution**
An organization in $\mathcal{Q}_{\mathrm{IV}}$ ($P > 0$, $W < 0$): productive for society, financed by participant sacrifice. Exists only through non-market motivation.

**Exchange institution**
An organization in $\mathcal{Q}_{\mathrm{I}}$ ($P > 0$, $W > 0$): creates value and compensates participants positively. The domain of classical political economy.

**Parasitic institution**
An organization in $\mathcal{Q}_{\mathrm{II}}$ or $\mathcal{Q}_{\mathrm{III}}$ ($P < 0$): net value destroyer. In $\mathcal{Q}_{\mathrm{II}}$ it pays members well (paying parasite); in $\mathcal{Q}_{\mathrm{III}}$ it also extracts from them (extractive parasite).

**Welfare decomposition**

The representation of $(P, W)$ as the sum of a social-welfare component along $(1, 1)^\top$ and a distributional component along $(1, -1)^\top$; see Proposition 3.5.

## The End